Back **Check your understanding: End of module 4**
Graded Assignment • 20 min

Due Jul 13, 11:59 PM IST

Your grade: 96%

Your latest: **96%** • Your highest: **96%** • To pass you need at least 80%. We keep your highest score.

Next item

1. Which of the following statements correctly describe properties or concepts related to transitive relations? 0.8 / 1 point

- [ ] A relation on any set is always transitive.
- [ ] Transitive relations must also be reflexive and symmetric.
- [x] For a set with a single element, every relation is transitive.

Nice work

Good job! A single element set trivially satisfies transitive properties.

- [ ] The transitive closure of a relation is the smallest transitive relation that contains the original relation.
- [x] If a relation is transitive, then for any elements a, b, and c, whenever aRb and bRc, then aRc.

Nice work

Correct! This is the definition of a transitive relation.

You didn't select all the correct answers

2. Which of the following is a property of an equivalence relation? 1 / 1 point

- ( ) Irreflexivity
- ( ) Antisymmetry
- ( ) Transitivity
- (•) Reflexivity

Nice work

Correct! Reflexivity is one of the properties of equivalence relations.

3. Given the adjacency matrix below, which property does the relation represented by this matrix have? 1 / 1 point

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

- ( ) The relation is symmetric
- ( ) The relation is anti-symmetric
- (•) The relation is reflexive
- ( ) The relation is transitive

Nice work

Correct! The diagonal of the matrix contains all 1s, indicating that the relation is reflexive.

4. Reflect on the mathematical skills developed throughout the course by identifying which of the following statements apply to the concept of asymptotic analysis in computing. 1 / 1 point

- [ ] Asymptotic analysis helps to determine the exact execution time of an algorithm in seconds.
- [x] Asymptotic analysis can be used to compare the relative efficiency of different algorithms.

Nice work

Great choice! The efficiency of algorithms in terms of growth rates and performance can be compared using asymptotic analysis.

- [x] Asymptotic analysis provides a way to evaluate the performance of an algorithm in terms of input size.

Nice work

Correct! Asymptotic analysis helps in understanding how the performance of an algorithm changes with varying input sizes.

- [ ] Asymptotic analysis is only relevant for analyzing non-recursive algorithms.
- [x] Asymptotic analysis uses Big O notation to describe upper bounds on time complexity.

Nice work

Well done! Big O notation is commonly used in asymptotic analysis to define upper bounds of an algorithm's time complexity.

5. Which of the following statements correctly describes a total order in a set? 1 / 1 point

- ( ) There exists at least one pair of elements in the set that is not comparable.
- ( ) Elements form a hierarchy, but not all elements need to be comparable.
- (•) Every pair of elements in the set is comparable.
- ( ) Every element is related to itself, but not necessarily to others.

Nice work

Correct! In a total order, any two elements in the set must be comparable, meaning one of them is either less than or equal to the other.